

The or That? Evaluating language models’ sensitivity to discourse structure in anaphora

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Abstract

We test whether language models (LMs) exhibit sensitivity to fine-grained discourse features that influence referential form choice in human language. Using a controlled paradigm from prior behavioral studies, we evaluate 19 LMs on their ability to distinguish between demonstratives (e.g., *that*) and definites (e.g., *the*) when referring back to previously mentioned entities, using expert-created stimuli across four languages (English, German, Turkish, Mandarin). In human data, demonstratives show selective licensing effects based on the presence or absence of contrastive noun phrases and discourse continuity. We find that only larger models replicate these effects reliably, with stronger correspondence to human judgments and higher item-level correlation. These results suggest that pragmatic constraints on reference are increasingly achievable at scale, and they introduce a new cross-linguistic testbed for probing the pragmatic competence of LMs.

1 Introduction

As language models (LMs) are increasingly deployed in real-world conversational settings, there has been growing interest in whether LMs *use* language pragmatically—for example, producing polite or relevant speech in the appropriate contexts (e.g., Hawkins et al., 2025; Zhao & Hawkins, 2025). But from the scientific perspective, it is also important to understand what LMs can learn about the pragmatic principles that govern natural language, simply through token prediction. Understanding the pragmatic generalizations encoded in LMs’ string probability distributions can reveal what aspects of pragmatic competence can arise through prediction over linguistic forms (Hu et al., 2023b).

Prior evaluations of LMs’ pragmatic competence have focused on phenomena such as presupposition, conversational implicature (e.g., Jeretic et al., 2020; Zheng et al., 2021; Ruis et al., 2023; Hu et al., 2023a; Sravanthi et al., 2024), and non-literal language understanding (Prystawski et al., 2023; Carenini et al., 2023; Oh et al., 2025; Tsvilodub et al., 2025). However, it has been relatively underexplored whether LMs track fine-grained informational-structural constraints in a way similar to humans. Some prior work has evaluated LMs’ sensitivity

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to high-level discourse structure (e.g., Beyer et al., 2021; Namuduri et al., 2025) or entity tracking in discourse (e.g., Sorodoc et al., 2020; Li et al., 2021; Loáiciga et al., 2022), but these studies have primarily focused on broad-coverage tasks in English, and have not performed fine-grained multilingual comparisons between LMs and humans.

Here, we focus on a pragmatic phenomenon that has been robustly demonstrated in humans across diverse languages: the contrast between *demonstrative* (e.g., “that”) and *definite* (e.g., “the”) descriptions when referring back to something previously mentioned in a conversation or text—a use known as *anaphora*. As an example, consider the following sentences:

- (1) { [1_{NP} A boy] / [2_{NP} A boy and a girl] } entered the classroom. (Context)
 a. The/That boy sat down in the front row. (Target – Same Situation)
 b. I had noticed the/that boy at a coffee shop yesterday. (Target – New Situation)

Recent behavioral studies across multiple languages have shown that demonstratives (e.g., “that boy”) and definites (e.g., “the boy”) are not interchangeable when referring back to something previously mentioned. While both forms can be used in anaphoric contexts, demonstratives are more selective in how they are used. Specifically, demonstratives are judged by speakers to be more natural when (i) only one prior referent is mentioned, and (ii) the context involves a change in situation—such as a temporal shift or a new speaker. In contrast, definites remain acceptable in a broader range of contexts as long as the referent is familiar and uniquely identifiable. This behavioral pattern has been robustly replicated across languages with different grammatical systems, and supports a more nuanced view of reference resolution models that account for discourse-level pragmatic constraints.

This is a valuable test bed for evaluating LMs’ pragmatic knowledge for several reasons. First, discourse phenomena are pervasive in natural language, and our focus on referential forms involving simple, high-frequency words (e.g., “the”, “that”) ensures that the evaluation remains ecologically valid and representative of real-world language use. Second, these phenomena are imbued in the structure of language itself, making it amenable to simple text-completion scoring methods, as opposed to multiple-choice or free-response scoring methods commonly used in pragmatics benchmarks. This feature allows us to evaluate arbitrarily small LMs with minimal auxiliary task demands (Hu & Frank, 2024).

Understanding LMs’ behavior in this domain can also inform ongoing research on how such preferences might operate in humans. If even small or simple LMs capture these preferences, this would suggest that these patterns can be easily gleaned from the statistics of language use. On the other hand, if these patterns only emerge in larger or more sophisticated LMs, this would suggest that human-like preferences involve more complex commonsense or discourse-related knowledge which is not immediately present in the linguistic signal. To foreshadow our results, we find that across four languages (English, German, Turkish, Mandarin), only *larger* LMs show human-like patterns of sensitivity to discourse-level structure, suggesting that scale plays a key role in supporting discourse-sensitive behavior.

2 Methods

Stimuli & human baseline. We used stimuli from prior work in four languages: English, German, Turkish, and Mandarin. For each language, the stimulus set followed a 2x2x2 design, forming 8 variants of 12 items. An example item from the English dataset is shown in Table 1. There were three critical manipulations: (1) **Same** or **New Situation** Targets; (2) **1** or **2 NPs** in the Context; and (3) **Definite** or **Demonstrative** description in the Targets. New situations always differed from same situations in being marked by both a new event participant (e.g. speaker or someone else) and a temporal change from the scenario in the context situation. Stimuli were also counterbalanced for both animacy and the syntactic positions (subject or object) of the target NPs. Note that Turkish and Mandarin both use bare nouns for definites, while English and German have determiners (e.g., “the”, “der”).

We compared LM behaviors to previously published human data using these stimuli. On each trial, humans saw a context sentence, as well as two target sentences, one with

Context sentence	Target sentence	Situation	NP	Position	Description type
A boy entered the classroom.	<u>The</u> boy sat down in the front row.	Same	1 NP	Subject	Definite
A boy entered the classroom.	<u>That</u> boy sat down in the front row.	Same	1 NP	Subject	Demonstrative
A boy entered the classroom.	I had noticed <u>the</u> boy at a coffee shop yesterday.	New	1 NP	Object	Definite
A boy entered the classroom.	I had noticed <u>that</u> boy at a coffee shop yesterday.	New	1 NP	Object	Demonstrative
A boy and a girl entered the classroom.	<u>The</u> boy sat down in the front row.	Same	2 NP	Subject	Definite
A boy and a girl entered the classroom.	<u>That</u> boy sat down in the front row.	Same	2 NP	Subject	Demonstrative
A boy and a girl entered the classroom.	I had noticed <u>the</u> boy at a coffee shop yesterday.	New	2 NP	Object	Definite
A boy and a girl entered the classroom.	I had noticed <u>that</u> boy at a coffee shop yesterday.	New	2 NP	Object	Demonstrative

Table 1: Example item from English dataset (Saha et al., 2023).

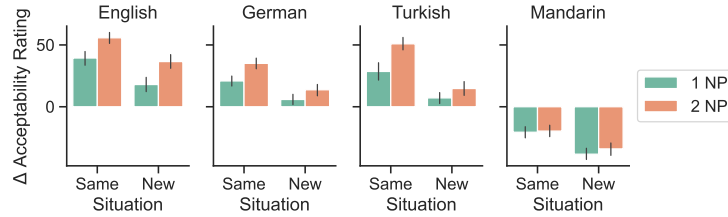


Figure 1: Human acceptability rating differences (Definite – Demonstrative), reproduced from Saha et al. (2023) (English, Turkish), Saha & Davidson (2024) (German), and Saha et al. (2024) (Mandarin).

the definite and one with the demonstrative. They then used sliders (0-100) to rate the acceptability of each target sentence, with endpoints labeled as “least natural” and “most natural”. The human data are shown in Figure 1, reproduced from Saha et al. (2023) (English, Turkish), Saha & Davidson (2024) (German), and Saha et al. (2024) (Mandarin). In English, German, and Turkish, there are two striking patterns: (1) a stronger preference for definites in the 2 NP condition, and (2) a weaker preference for definites in the New Situation condition. However, Mandarin diverges from this pattern. In Mandarin, demonstratives are consistently preferred over definites (expressed with bare nouns, as Mandarin lacks definite determiners) across all conditions, and the preference is stronger in New Situation condition. Note that Turkish also expresses definites with bare nouns, but still exhibits a similar pattern to English and German. This cross-linguistic variation underscores the need for typologically grounded evaluation when interpreting model behavior across languages.

LM evaluation. We evaluated 19 unique pretrained LMs (see Table 2, Appendix A) by computing log-probability differences between Definite and Demonstrative continuations. For *autoregressive* LMs, we measured the probability of the target sentence given the context sentence. For *bidirectional* LMs, we masked the determiner and measured the model’s preference given the preceding and following context.¹ We compared human and model *relative preferences* for the definite form: that is, we compared *differences* in human acceptability ratings to *differences* in LM log-probabilities (Definite – Demonstrative).

We also note that the stimuli used for evaluation were published after the tested LMs’ training data cutoff dates, so there is minimal risk of data contamination.

¹Note that only the English and German stimuli were compatible with bidirectional scoring, as the definite forms in Mandarin and Turkish are bare nouns, so there is no content to mask out.

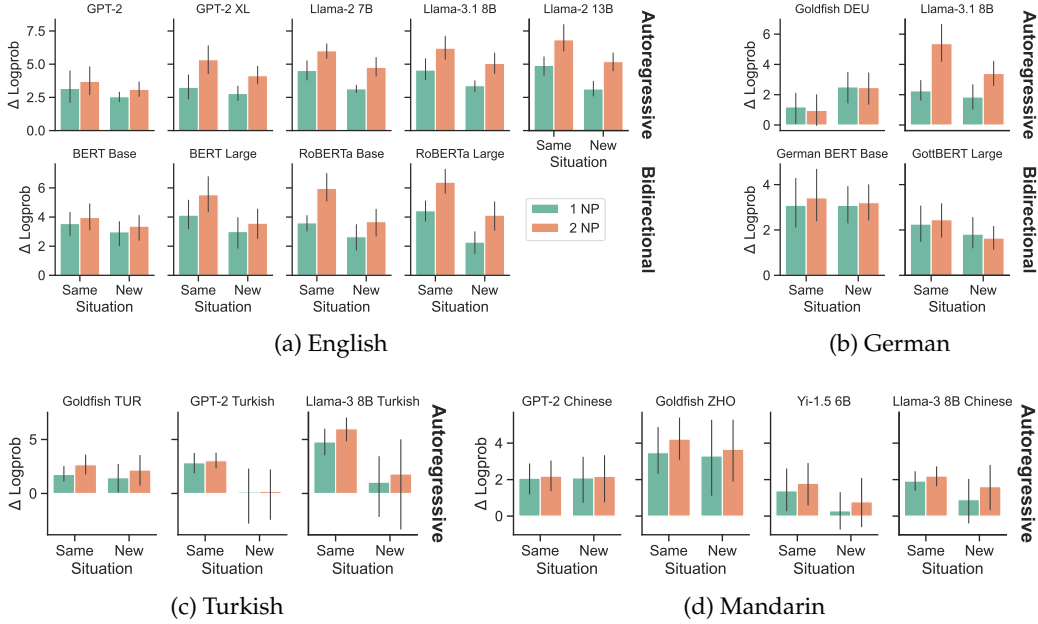


Figure 2: LM log-probability differences (Definite – Demonstrative).

3 Results & Discussion

Only larger LMs capture human-like behavioral patterns. Figure 2 shows the main results for the tested languages and models. Each bar shows the mean difference in log-probability assigned to the Definite versus Demonstrative variants, averaged across items. We first focus on the English results (Figure 2a). For smaller models, such as GPT-2 and BERT Base, we find that log-probability differences capture only *one* of the signature patterns seen in the human data: an overall stronger preference for definites in the 2NP case. In contrast, the larger and more advanced models (e.g., Llama, RoBERTa) show *both* of the signature patterns seen in the human data: the preference for the definite form is (1) substantially higher in the 2 NP condition, and (2) overall lower in the New Situation condition. These trends were supported by one-sided *t*-tests.

Turning to the other languages, we find qualitatively similar patterns for the autoregressive German LMs (Figure 2b), and the Turkish and Mandarin LMs (Figures 2c and 2d, respectively). In Mandarin, LMs show a consistent preference for definite continuations across all conditions, unlike the consistent preference for demonstratives in human ratings. This divergence likely reflects the nature of Mandarin definites, which are realized as bare nouns owing to the lack of a definite determiner in the language, coupled with LM bias toward shorter continuations. Despite this global preference, LMs still capture key human-like trends: we find no increased preference for definites in the 2 NP condition, and a weaker preference for definites in the New Situation condition. Notably, Turkish patterns with English and German despite lacking a definite determiner (like Mandarin), showing that overt marking is not required for discourse-sensitive effects to emerge. This highlights how typological differences and model biases can interact in language-specific ways.

Humans and larger LMs are sensitive to similar input features. Next, we investigated whether humans and models are sensitive to similar properties of the stimuli. Figure 3 shows the by-item human and model relative preferences for the Definite continuation, for the English autoregressive models. While each LM achieves a significant correlation with the human data, there is a clear effect of model size (e.g., Llama-2 13B Pearson $r = 0.66$). We find similar effects of size for the English bidirectional models (e.g., BERT Base $r = 0.17$; BERT Large $r = 0.46$), as well as the other three languages, although the correlations are

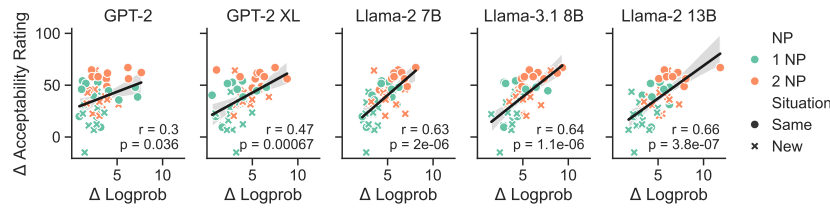


Figure 3: Item-level relationship between human acceptability rating differences (y-axis) and model log-probability differences (x-axis), for the English autoregressive models.

overall weaker (< 0.4) for the German bidirectional models and the Turkish models. The Pearson r values achieved by the full set of models are shown in Figure 4, Appendix B.

General discussion. The fact that only larger models reliably captured human-like preference patterns suggests that pragmatic constraints are not easily learnable from surface-level statistics alone. Our study illustrates how even basic aspects of language—such as preferences for when to use words like “the” and “that”—can reflect sophisticated pragmatic knowledge, and meaningfully differentiate between neural LMs. An exciting direction for future work would be to test the effect of monolingual vs. multilingual training, and compare the strategies that humans and LMs use to compute discourse-sensitive preferences.

Acknowledgments

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A Models

Name	Huggingface ID	Language	Type	Size (B)
GPT-2	gpt2	English	Autoregressive	0.124
GPT-2 XL	gpt2-xl	English	Autoregressive	1.5
Llama-2 7B	meta-llama/Llama-2-7b-hf	English	Autoregressive	7
Llama-3.1 8B	meta-llama/Llama-3.1-8B	Multilingual (English)	Autoregressive	8
Llama-2 13B	meta-llama/Llama-2-13b-hf	English	Autoregressive	13
BERT Base	google-bert/bert-base-cased	English	Bidirectional	0.110
BERT Large	google-bert/bert-large-cased	English	Bidirectional	0.340
RoBERTa Base	FacebookAI/roberta-base	English	Bidirectional	0.125
RoBERTa Large	FacebookAI/roberta-large	English	Bidirectional	0.360
Goldfish DEU	goldfish-models/deu_latn.1000mb	German	Autoregressive	0.125
Llama-3.1 8B	meta-llama/Llama-3.1-8B	Multilingual (German)	Autoregressive	8
German BERT Base	google-bert/bert-base-german-cased	German	Bidirectional	0.110
GottBERT Large	TUM/GottBERT_large	German	Bidirectional	0.355
Goldfish TUR	goldfish-models/tur_latn.1000mb	Turkish	Autoregressive	0.125
GPT-2 Turkish	ytu-ce-cosmos/turkish-gpt2	Turkish	Autoregressive	0.125
Llama-3 8B Turkish	ytu-ce-cosmos/Turkish-Llama-8b-v0.1	Multilingual (Turkish)	Autoregressive	8
GPT-2 Chinese	uer/gpt2-chinese-cluecorpussmall	Mandarin	Autoregressive	0.125
Goldfish ZHO	goldfish-models/zho_hans.1000mb	Mandarin	Autoregressive	0.125
Yi-1.5 6B	01-ai/Yi-1.5-6B	Multilingual (Mandarin)	Autoregressive	6
Llama-3 8B Chinese	hf1/llama-3-chinese-8b	Multilingual (Mandarin)	Autoregressive	8

Table 2: Models evaluated in our experiments. Models are trained on monolingual data unless otherwise noted, with the evaluation language labeled in parentheses.

B Item-level correlations

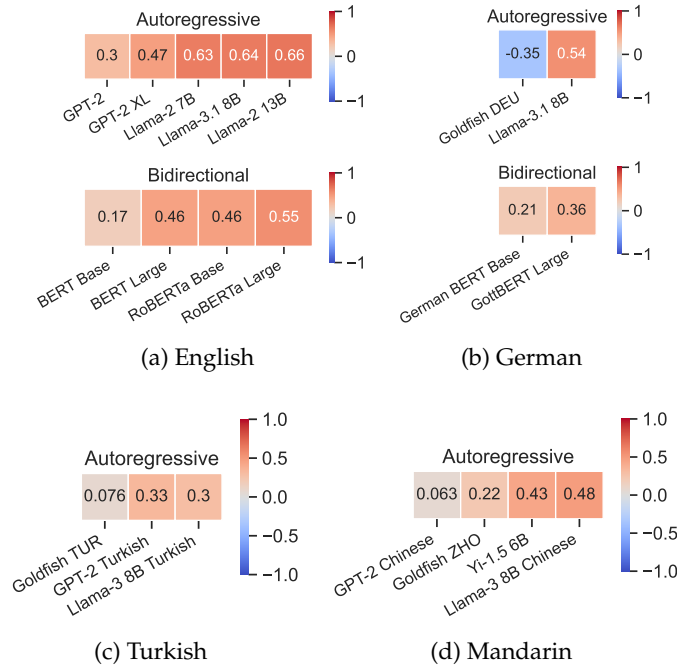


Figure 4: Pearson r correlations between item-level human acceptability rating differences and model log-probability differences (Definite–Demonstrative).